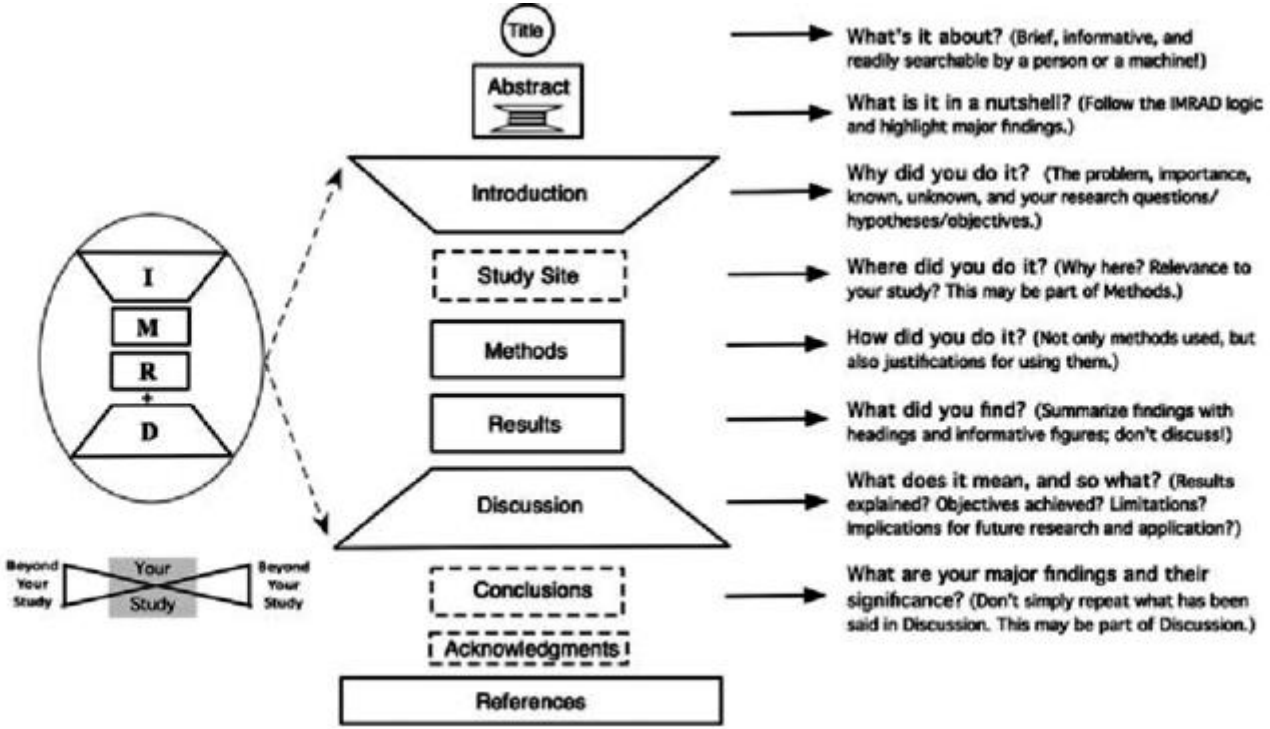
ITMO

IMRAD.
I - Introduction

IMRaD structure

İTMO



The figure above is symmetrical: many of the things that are in the Introduction occur — often in reverse order — in the Discussion.

- At the start of the Introduction you write an **opening sentence** which enables your reader to ‘get in’ or start your paper/thesis; at the end of the Discussion you ‘get out’ by finding an appropriate way to end.
- During the Introduction you generally **present past or current research** and/or knowledge about the topic; in the Discussion you show how your study contributes to that research or advances that knowledge.
- At some point in the Introduction you **state the specific problem** your study focuses on or **identify a gap** in the existing research that will be filled in your paper; in the Discussion you discuss to what extent you have resolved that problem.

In the Introduction you start out by being fairly general and gradually narrow your focus towards your own study, and the opposite is true in the Discussion.

- You start with a fairly general item of information, one that many of your readers already know. This represents **a ‘meeting place’ fact** from which all readers can start together, after which you can move on to more specific information.

- Always show your readers **the general picture** before you proceed to the details: show them the wall before you start to talk about the bricks.

- As a general rule it is better to **provide slightly too much** background information than slightly too little**.**

The key message for all the sections is as follows:

Evidence-based guidance can be obtained by studying the structure and language of successfully-published papers.

To keep the review in the Introduction on track, **plan and organise** the research contributions according to an organisational pattern.

Here are three possible patterns:

- ***General–to–specific***

Writers frequently start with general research in the field, possibly a review article, and gradually move to research that is closer to the problem or gap dealt with in the present study. This is the most common pattern.

- ***Different approaches/theories/model****s*

In some papers, studies are grouped according to the approach or method they used. Grouping similar studies together helps you avoid the 'tennis match' effect where you go backwards and forwards beginning each sentence with *However* or *On the other hand.*

- ***Chronological order***

This may be appropriate, for example, if the development of your field is related to political decisions and laws such as those relating to pollution or fracking.

GENERIC INTRODUCTION MODEL

1	ESTABLISH THE IMPORTANCE OF THE TOPIC/FIELD PROVIDE BACKGROUND FACTUAL INFORMATION PRESENT THE GENERAL PROBLEM AREA/CURRENT RESEARCH FOCUS
2	PRESENT PREVIOUS AND/OR CURRENT RESEARCH AND CONTRIBUTIONS: the research ‘map’
3	LOCATE A GAP IN THE RESEARCH DESCRIBE THE PROBLEM YOU WILL ADDRESS PRESENT YOUR MOTIVATION AND/OR HYPOTHESIS IDENTIFY A RESEARCH OPPORTUNITY
4	DESCRIBE THE PRESENT PAPER, sometimes mentioning aims/ results/methods/conclusions

The Introduction model

In Sentence 1, the writer	establishes the importance of this research topic.
In Sentence 2, the writer	provides background factual information for the reader.
In Sentence 3, the writer	does the same as in Sentences 1 and 2, but in more detail, using citation references to support the background facts and the claim of importance.
In Sentence 4, the writer	describes the general problem area or the current research focus of the field.
In Sentence 5, the writer	extends the problem area/current focus of the field.
In Sentence 6, the writer	links the general problem area to published research and refers to other studies.
In Sentence 7, the writer	mentions a key research study in this area.
In Sentence 8, the writer	identifies a problem or gap in the research.
In Sentence 9, the writer	describes the present paper or study.
In Sentence 10, the writer	outlines the method reported in the paper.
In Sentence 11, the writer	announces the main result, using language that explicitly identifies the achievement of the study.

Identifying the model components

In an example below, underline the words and phrases that help identify the model components

The interaction between metallic uranium surfaces and water vapour is considered to be most important in regard to the environmental corrosion of the metal. Numerous studies have examined the initial stages of these interactions [1−7]. However, there have been discrepancies in the published data describing kinetic laws, pressure dependence of the reaction rate constant and activation energies. The precise mechanism for uranium corrosion is not entirely clear with different mechanisms proposed arising from the results of the undertaking studies [1−7]. Existing discrepancies in the published data may, in part, be related to differences in the provenance and purity of the metal used by different groups. The reactivity of impurity species such as carbide, may have affected recorded data. This work aims to provide data for an improved understanding of the role of impurity phases in the uranium−water reaction, samples of uranium containing 600 ppm carbon were analysed during and after exposure to water vapour at 19 mbar pressure, in an environmental scanning electron microscope (ESEM). Samples were analysed using secondary ion mass spectrometry (SIMS), focused ion beam (FIB) imaging and sectioning and transmission electron microscopy (TEM) with X−ray diffraction (micro−XRD).

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Language for the Introduction

ESTABLISHING SIGNIFICANCE (usually in the first sentence)

- The status and function of the port has evolved **rapidly in recent years**.
- Since the discovery of the first isolated graphene layer, **many** chemical approaches have been developed.
- The **increasing** pressure to eliminate lead has stimulated **great interest** in the search for lead−free solders.
- Fourier transforms are **widely used** in image processing to characterise textures in images.
- The vanadium redox flow battery (VRFB) is an **emerging** energy storage technology.

Your examples?

Language for the Introduction

REFERRING TO RESEARCH ACTIVITY AND CONTRIBUTIONS IN PREVIOUS AND CURRENT RESEARCH

- The effect of keratin on the mechanical properties of copolymer systems **was demonstrated** more than two years ago.
- Minelli *et al.* **describe** the use of experts' opinions and empirical evidence.
- NDE systems **have been applied** to the inspection of pipelines.
- Several authors **have developed** compensation techniques for temperature changes.
- The advantages and limitations of each of these techniques **have been evaluated** by different authors.

Your examples?

Language for the Introduction

IDENTIFYNG GAP

- The fact that these galaxies are so massive may make them **ideal candidates** for protocluster searches.
- **Little attention has been paid** to the selection of an alternative biocompatible material.
- How propofol acts on neural circuits to produce unconsciousness **remains unclear**.
- The main **drawback** of this approach is that it is **unable** to provide quantitative information.
- **Little is known** about these early differentiating cells and **it is not clear** whether this initial Oct4 expressing population differ from undifferentiated hESCs.

Your examples?

Language for the Introduction

THE PRESENT WORK

- **Our study focuses on**…
- To address this question, **we investigated**…
- Section V **provides several relevant** conclusions obtained from **the study reported in this paper**.
- **The main objective of this study** was **to describe and examine**…
- **In this paper we present** a **robust** method for…
- **The proposed method** combines GWAS data with information from bioinformatics databases in a **coherent and reproducible** way.
- **In this study**, we **carefully identified and isolated**…

Your examples?

Language for the Introduction

Put the following sentences in the correct order.

1. The essay which follows gives a brief history of prepositional theory and discusses two major teaching strategies from a cognitive linguistic perspective.
2. These small connecting words do not necessarily exist in other languages, or may not have exactly the same meanings.
3. The use of prepositions in English has always been problematic for language learners.
4. This makes teaching of this area very difficult, and research (Lindstromberg 1998; Brala 2002) indicates that no single method has yet proved successful.

Language for the Introduction

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THANK YOU FOR YOUR TIME!

ITMO more than a UNIVERSITY